# Test-procedure choice for the biomarker-treatment interaction in aggregated N-of-1 trials: dichotomization, classical RM-ANOVA, mixed-effects, and generalized estimating equations

pmsimstats team

October 28, 2026

## Contents

**1 Abstract** **2**

**2 Introduction** **3**

2.1 The question . . . . . 3

2.2 Why the interaction and the N-of-1 setting . . . . . 4

2.3 What the literature does and does not do . . . . . 4

2.4 What this paper contributes . . . . . 5

**3 Closed-form strict RM-ANOVA test of the biomarker-treatment interaction** **6**

3.1 Notation and design . . . . . 6

3.2 Sums of squares and the interaction $F$-statistic . . . . . 7

3.3 Relation to the linear-mixed Wald test . . . . . 8

**4 Simulation design** **10**

4.1 Aims . . . . . 10

4.2 Trial design . . . . . 10

4.3 Data-generating mechanism . . . . . 11

4.4 Estimands . . . . . 11

4.5 Methods . . . . . 12

4.6 Performance measures . . . . . 13

4.7 Deviations from the working protocol . . . . . 13

4.8 Execution . . . . . 14

**5 Results** **15**
5.1 Type I error and power . . . 15
5.2 The cost of dichotomization . . . 16
5.3 GEE calibration . . . 16
5.4 Standard-error calibration . . . 17
5.5 The conservatism of the model-based LME . . . 18
5.6 LME plus CR2 arm . . . 18

**6 Recommendation** **19**

**7 Discussion** **20**
7.1 What the study settles . . . 20
7.2 Limitations . . . 20
7.3 Future work . . . 21

**8 Conclusions** **21**

**9 Glossary** **21**

**10 Conflict of interest** **22**

**11 Funding** **22**

**12 Data availability statement** **22**

**13 Reproducibility** **23**

**14 References** **23**

# 1 Abstract

**Background.** Inference about the biomarker-treatment interaction in aggregated N-of-1 trials depends on the test procedure. The classical repeated-measures ANOVA (RM-ANOVA) requires a dichotomized biomarker. Mixed-effects models and generalized estimating equations (GEE) do not. We separate the cost of dichotomization from the cost of the test procedure.

**Methods.** We derive the closed-form strict RM-ANOVA $F$-statistic for the biomarker-stratified treatment interaction in a balanced split-plot design. We show that it is a two-sample $t$-test on within-participant phase differences. We then ran a Monte Carlo study reported under the ADEMP framework[1]. Six procedures were compared. Strict RM-ANOVA

used a median-split biomarker. A within-participant contrast kept the biomarker continuous and otherwise matched RM-ANOVA. A linear mixed model (LME) used continuous-time AR(1) residuals, once with model-based inference and once with CR2 cluster-robust inference. Two GEE analyses used the Liang-Zeger and the Mancl-DeRouen sandwich. The design was a two-path open-label plus blinded discontinuation schedule with $N \in \{25, 35, 70, 100\}$ and an interaction coefficient $\beta_{bm} \in \{0, 0.45\}$. Each power cell had 1000 replicates and each null cell 2000.

**Results.** Dichotomization accounted for 0.08 to 0.21 of absolute power. Among procedures with nominal or conservative size, the test procedure accounted for at most 0.07. The observed loss matched Cohen's $2/\pi$ efficiency factor for a median split. The Liang-Zeger GEE rejected at 7.4 to 9.7 percent under the null. The Mancl-DeRouen sandwich with a $t$ reference on $K - 5$ degrees of freedom rejected at 4.7 to 5.9 percent. Its power was 10 to 15 points below the uncorrected GEE at $N \leq 35$. The LME with model-based inference rejected at 2.1 to 3.4 percent. Its standard error exceeded the empirical standard error by 10 to 16 percent. The same fit with CR2 inference rejected at 4.3 to 5.9 percent and gained 0.5 to 3.0 points of power.

**Conclusions.** Keeping the biomarker continuous is the most consequential choice. The LME's conservatism is a working-covariance calibration defect. Degrees-of-freedom corrections cannot repair it. The cluster-robust CR2 standard error used by the companion papers repairs it here. Uncorrected GEE should not be used at these sample sizes.

# 2 Introduction

## 2.1 The question

An analyst testing a biomarker-treatment interaction in an aggregated N-of-1 trial chooses a test procedure. The classical choice is a repeated-measures ANOVA (RM-ANOVA) on a biomarker split into high and low strata. The modern choices are a linear mixed-effects model (LME) or generalized estimating equations (GEE) with the biomarker kept continuous. The two routes differ in two ways at once. The classical route dichotomizes the biomarker. The modern route models the within-participant covariance. A direct comparison of RM-ANOVA against a mixed model confounds the two.

This paper separates them. We add a fourth procedure, a within-participant contrast regressed on the continuous biomarker. It differs from strict RM-ANOVA only in keeping the biomarker continuous. It differs from the mixed model only in collapsing each phase to its mean. The step from RM-ANOVA to the contrast measures the cost of dichotomization. The step from the contrast to the mixed model measures the cost of the test procedure. We report both steps across four sample sizes.

We also report the small-sample calibration of the four continuous-biomarker procedures. GEE with the Liang-Zeger sandwich is known to over-reject with few clusters. Bias-corrected sandwiches exist, and the choice of reference distribution matters as much as the correction.

The LME with model-based inference can be conservative when its working covariance is misspecified. We tabulate size, power, coverage, and the calibration ratio $\kappa$ (mean model standard error over the empirical standard deviation of the estimates) for each arm and cell.

## 2.2 Why the interaction and the N-of-1 setting

The biomarker-treatment interaction is harder to detect than the average treatment effect. The PATH framework[2,3] distinguishes effect modeling from risk modeling. The interaction is estimated from within-participant treatment contrasts, and its precision depends on the number and placement of off-drug visits. A test procedure that loses information on the interaction will miss predictive-biomarker signal at sample sizes that are adequate for the main effect.

Within-participant residuals in chronic-condition trials are serially correlated on a timescale set by pharmacokinetics, natural-history drift, and measurement reactivity. A continuous-time AR(1) residual correlation represents this. It is not spherical in the sense required by a classical analysis that keeps every visit as a repeated level.[4] report through simulation that uncorrected RM-ANOVA inflates Type I error under sphericity violations. They also report that the Greenhouse-Geisser-corrected RM-ANOVA and the multilevel model perform comparably, and that the multilevel model is not uniformly safer.[5] extend this with finer epsilon resolution. The classical procedure we evaluate pre-averages each phase to one value, so it has a two-level within-participant factor and sphericity holds by construction. The serial correlation affects it only through the variance of the phase difference.

## 2.3 What the literature does and does not do

Most published N-of-1 analyses ignore serial correlation.[6] report that 83.8 percent of the studies they catalog do so.[7] and[8] reach similar conclusions in neurology and behavioral medicine, and[9] in rare-disease applications. The CENT reporting standards[10,11] do not address the analytic-model decision, and[12] rank it as the field's highest-priority methodological problem.

The cost of a median split is a classical result.[13] gave the efficiency loss of dichotomizing a normal predictor at the median as about 36 percent, the factor $2/\pi$ on the squared correlation.[14] documented the practice and its consequences at length.[15] showed the power loss and residual confounding in regression. The present paper observes that loss once more, in the N-of-1 interaction setting, and compares it with the $2/\pi$ prediction.

Test-procedure comparisons for longitudinal data are not new.[16] report that RM-ANOVA is less powerful than GEE and mixed models for continuous longitudinal endpoints. The MMRM literature[17,18] addresses parallel-group trials with monotone missingness. The GEE of[19] over-rejects with few clusters, and several bias-corrected sandwiches address this[20–23].[24] proposed $t$ and $F$ references that account for the variability of the sandwich.[25] compared the corrections and concluded that the Wald $z$ test should be avoided with few clusters even with a bias-corrected sandwich, and that a $t$ reference is needed.[26] report that in stepped-wedge

trials with fewer than six units only 6.6 percent of analyses use a small-sample correction. None of these papers addresses the biomarker-interaction estimand in an N-of-1 design.

Design optimization has been studied separately from test-procedure choice.[27] give power computations for $(AB)^k$ single-case designs.[28] document the effect of serial correlation on randomization-test power.[29] and[30] show that replicate crossover designs identify the patient-by-treatment interaction variance.[31] compare design families for the biomarker-interaction question with one test procedure.[32] compare aggregated N-of-1 trials against parallel and crossover designs with one mixed model.[33] and[34] review the design literature. Run-in selection is a separate controversy:[35] show that response-based exclusion during a placebo run-in biases validity, and the open-label titration used here avoids it.

## 2.4 What this paper contributes

The paper makes three contributions.

First, we derive the closed-form strict RM-ANOVA $F$-statistic for the biomarker-stratified treatment interaction in a balanced split-plot design (Section 3). We show that it is a two-sample $t$-test on within-participant phase differences and that the reduction is exact for any within-participant covariance once each phase is averaged. We relate it to the LME Wald test.

Second, we run a Monte Carlo study that separates the cost of dichotomization from the cost of the test procedure (Sections 4 and 5). The continuous-biomarker contrast is the instrument. The study uses the pmsimstats package data-generating process under the mean-shift encoding of the interaction, at a two-path open-label plus blinded discontinuation design with no carryover.

Third, we tabulate the small-sample calibration of the four continuous-biomarker procedures and relate it to the inference the companion papers recommend (Sections 5 and 6). The LME arm evaluated here is the series' historical analysis with model-based inference. Its conservatism is a calibration defect of the working covariance, and the companion papers[36,37] remedy it with a cluster-robust standard error. We add that arm and report it beside the model-based LME.

**Notation for the interaction parameter.** The data-generating process shifts the drug-response mean by the centered biomarker. We write the interaction parameter as $\beta_{bm}$, a dimensionless multiplier on the standardized biomarker. The companion architecture manuscript uses $c_{bm}$ for the correlation encoding. The two are calibrated so that 0.45 denotes a matched interaction strength, but the symbols are not interchangeable. The biomarker is standardized (mean 0, SD 1) throughout and carries no clinical units.

The remainder of the paper is organized as follows. Section 3 gives the derivation, adapted from `docs/legacy/26-rm-anova-interaction-test.md`. Section 4 gives the simulation design. Section 5 reports the results. Section 6 gives the recommendation. Section 7 discusses limitations and future work.

# 3 Closed-form strict RM-ANOVA test of the biomarker-treatment interaction

This section derives the strict RM-ANOVA $F$-statistic for the biomarker-treatment interaction in a balanced split-plot design and relates it to the Wald $t$-statistic on the corresponding LME coefficient. The derivation specializes the classical split-plot analysis[38] and the crossover treatment of within-subject contrasts[39] to the simplest design that supports the test: two biomarker strata, two treatment phases, and $T$ within-phase timepoints. Three facts anchor the rest of the paper.

1. Under balance, the strict RM-ANOVA test of the interaction is a two-sample $t$-test on the within-participant phase differences with $2(n-1)$ degrees of freedom. The reduction is exact for any within-participant covariance once each phase is averaged.
2. The closed form gives a non-central $F$ power formula.
3. The closed form assumes a categorical biomarker. The present study exercises that assumption and no other.

## 3.1 Notation and design

We consider the following elements.

- Two biomarker strata $i \in \{1, 2\}$, formed by dichotomizing the continuous biomarker at a cut-point. Strict RM-ANOVA requires categorical predictors.
- $n$ participants per stratum, indexed $l \in \{1, \dots, n\}$.
- Two treatment phases $j \in \{1, 2\}$, drug and placebo. Every participant passes through both.
- $T$ within-phase timepoints, indexed $k \in \{1, \dots, T\}$. The analyst may pre-average them to one value per participant per phase.

The biomarker stratum is between participants. The phase and the timepoints are within participants. This is the split-plot repeated-measures design[38].

We write $Y_{ijlk}$ for the outcome at stratum $i$, phase $j$, participant $l$, timepoint $k$. The classical split-plot model is

$$\begin{aligned} Y_{ijlk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \pi_{l(i)} \\ + \tau_k + (\alpha\tau)_{ik} + (\beta\tau)_{jk} + (\alpha\beta\tau)_{ijk} + e_{ijlk}, \end{aligned}$$

with sum-to-zero constraints, a random participant effect $\pi_{l(i)} \sim N(0, \sigma_\pi^2)$ nested in stratum, and error $e_{ijlk} \sim N(0, \sigma_e^2)$. The model implies compound symmetry of the within-participant covariance. The $F$-tests of within-participant effects require only sphericity, the equality of the variances of all pairwise differences among the levels of a within-participant factor. Compound symmetry implies sphericity. A within-participant factor with two levels is always spherical.

### 3.2 Sums of squares and the interaction $F$-statistic

The total sum of squares splits into between- and within-participant components:

$$SS_{\text{tot}} = SS_{\text{between}} + SS_{\text{within}}. \tag{1}$$

The between component decomposes into $SS_A$ (stratum) and $SS_{S(A)}$ (participant within stratum):

$$SS_A = bnT \sum_{i=1}^{a} (\bar{Y}_{i\cdots} - \bar{Y}_{\cdots})^2, \quad SS_{S(A)} = bT \sum_{i=1}^{a} \sum_{l=1}^{n} (\bar{Y}_{i\cdot l\cdot} - \bar{Y}_{i\cdots})^2. \tag{2}$$

The within component decomposes into $SS_B$ (phase), $SS_{AB}$ (the interaction), and the within error $SS_{B\times S(A)}$:

$$SS_{AB} = nT \sum_{i,j} (\bar{Y}_{ij\cdot\cdot} - \bar{Y}_{i\cdots} - \bar{Y}_{\cdot j\cdot\cdot} + \bar{Y}_{\cdots})^2, \tag{3}$$

$$SS_{B\times S(A)} = T \sum_{i,j,l} (\bar{Y}_{ijl\cdot} - \bar{Y}_{i\cdot l\cdot} - \bar{Y}_{ij\cdot\cdot} + \bar{Y}_{i\cdots})^2. \tag{4}$$

The degrees of freedom are $df_A = a-1$, $df_{S(A)} = a(n-1)$, $df_B = b-1$, $df_{AB} = (a-1)(b-1)$, and $df_{B\times S(A)} = a(n-1)(b-1)$. The interaction test is

$$F_{AB} = \frac{MS_{AB}}{MS_{B\times S(A)}} = \frac{SS_{AB}/df_{AB}}{SS_{B\times S(A)}/df_{B\times S(A)}}. \tag{5}$$

Under $H_0 : (\alpha\beta)_{ij} \equiv 0$ and sphericity, $F_{AB} \sim F(df_{AB}, df_{B\times S(A)})$.

#### 3.2.1 The $2 \times 2$ split-plot case

We specialize to $a = b = 2$ and $T = 1$, so that the analyst pre-averages within-phase observations to one value per participant per phase. The sample interaction contrast is

$$\hat{L} = (\bar{Y}_{11\cdot} - \bar{Y}_{12\cdot}) - (\bar{Y}_{21\cdot} - \bar{Y}_{22\cdot}). \tag{6}$$

$\hat{L}$ is the difference of within-participant treatment differences across strata. Under the sum-to-zero parameterization, $\widehat{(\alpha\beta)}_{11} = \hat{L}/4$, so $SS_{AB} = nT\,\hat{L}^2/4$. With $df_{AB} = 1$ this is the interaction mean square. Under sphericity and balance, $MS_{B\times S(A)} = \hat{\sigma}_e^2$. Substituting into equation (5),

$$F_{AB} = \frac{nT\,\hat{L}^2}{4\,\hat{\sigma}_e^2}, \qquad df_1 = 1, \quad df_2 = 2(n-1). \tag{7}$$

We call equation (7) the strict RM-ANOVA $F$ throughout.

### 3.2.2 Equivalence to a within-participant $t$-test

Define $\Delta_{il} = \bar{Y}_{i1l\cdot} - \bar{Y}_{i2l\cdot}$, the within-participant phase difference for participant $l$ in stratum $i$. Then

$$\hat{L} = \bar{\Delta}_1 - \bar{\Delta}_2, \qquad \bar{\Delta}_i = \frac{1}{n}\sum_{l=1}^{n} \Delta_{il}. \tag{8}$$

Under the model, the $\Delta_{il}$ are i.i.d. within stratum with $\mathrm{Var}(\Delta_{il}) = 2\sigma_e^2/T$. The participant effect cancels in the difference. For any within-participant covariance shared by all participants, AR(1) included, the $\Delta_{il}$ remain i.i.d. normal within stratum with a common variance. Only the value of that variance changes. The variance of $\hat{L}$ is

$$\mathrm{Var}(\hat{L}) = \frac{2 \cdot 2\sigma_e^2}{nT} = \frac{4\sigma_e^2}{nT}, \tag{9}$$

and the $t$-statistic is

$$t_{AB} = \frac{\hat{L}}{\sqrt{4\hat{\sigma}_e^2/(nT)}} = \sqrt{\frac{nT}{4}} \cdot \frac{\hat{L}}{\hat{\sigma}_e}. \tag{10}$$

Then $t_{AB}^2 = F_{AB}$, which recovers equation (7). The strict RM-ANOVA $F$-test for the interaction is a two-sample $t$-test on the within-participant phase differences with $2(n-1)$ degrees of freedom. The identity is standard in the split-plot literature[38].

The pooled $t$-test needs only that the $\Delta_{il}$ be i.i.d. normal with a common variance in the two strata. The identity and the exactness at $2(n-1)$ degrees of freedom therefore hold for any within-participant covariance once each phase is averaged. Two qualifications apply to the executed study. First, the two randomization paths have different numbers of off-drug visits (3 and 4, counting baseline), so the contrast variance differs by path and the pooled $t$ is approximate. Second, with $N$ odd the median split gives unequal strata (13 and 12 at $N = 25$), so the $2(n-1)$ statement assumes balance. The observed null rejection rates of the arm (Section 5) show that neither approximation matters here.

For $a$ strata, $b$ phases, $n$ participants per stratum, and $T$ timepoints, equation (5) generalizes with $df_1 = (a-1)(b-1)$ and $df_2 = a(n-1)(b-1)$. With the time factor retained, the test of whether the biomarker modifies the treatment effect across time is the three-way interaction, with $F_{AB\tau} = MS_{AB\tau}/MS_{B\tau\times S(A)}$, $df_1 = (a-1)(b-1)(T-1)$ and $df_2 = a(n-1)(b-1)(T-1)$.

## 3.3 Relation to the linear-mixed Wald test

The strict RM-ANOVA $F$ of equation (7) and the `nlme::lme` Wald $t$-statistic on the `bm:Dbc` coefficient are asymptotically equivalent under three conditions: (i) the biomarker is di-

chotomized at the cut-point that defines the strata, (ii) the within-participant covariance is compound-symmetric, and (iii) `Dbc` is the binary on-drug indicator. When all three hold, the two tests reach the same coefficient and the same standard error up to small-sample factors. When any one fails, the mixed model uses information the strict RM-ANOVA discards. The continuous biomarker contributes within-stratum variation. The AR(1) covariance is modeled. The exposure-decay predictor uses the off-drug trajectory.

The sphericity condition binds for the classical analysis that keeps the $T$ within-phase timepoints as repeated levels. The pre-averaged $2\times2$ form has a two-level within-participant factor and is spherical for any covariance. When sphericity fails, the nominal $F$-distribution is no longer exact under $H_0$ and the test over-rejects. Mauchly's test[40] tests sphericity but is uninformative about the direction of departure. Box's $\epsilon$ measures the departure, with $\epsilon = 1$ under sphericity and $\epsilon \geq 1/(T-1)$ at maximum departure. The Greenhouse-Geisser correction[41] estimates $\hat{\epsilon}$ and scales both degrees of freedom by it. The Huynh-Feldt correction[42] is less conservative and is recommended when $\hat{\epsilon} > 0.75$[4,5].

The pmsimstats DGP imposes AR(1) correlation within each component. A classical analysis that keeps the full visit series as within-participant levels therefore over-rejects without a correction. The project's audit (`docs/legacy/06-ar1-residual-correlation.tex`) fitted a mixed model with no residual-correlation structure to AR(1) data at eight post-baseline visits and found Type I error of 9 to 17 percent across the crossover and open-label designs at $N = 35$ and 70 (crossover 0.17 and 0.13, open-label 0.13 and 0.09). Switching to `corCAR1` returned it to nominal. That figure is for the unstructured fit, not for the strict RM-ANOVA arm evaluated here. The arm evaluated here pre-averages each phase, so its Type I error is at nominal (Section 5) and its operative cost is dichotomization.

Where a reviewer requests RM-ANOVA results, three responses are defensible: (a) point to the LME as a continuous-time member of the RM-ANOVA family, (b) report the strict RM-ANOVA $F$ of equation (7) as a sensitivity comparator, or (c) report both with the power-loss decomposition. The simulation study operationalizes (c).

### 3.3.1 Power calculation under the closed form

Under the alternative with $\mathbb{E}[\hat{L}] = \Delta_0$, the statistic $F_{AB}$ follows a non-central $F$ distribution with non-centrality parameter

$$\lambda = \frac{nT\,\Delta_0^2}{4\sigma_e^2}, \tag{11}$$

where $\Delta_0$ is the true interaction contrast on the outcome scale. Power at level $\alpha$ is

$$1 - \beta = \Pr\Big[\, F_{AB} > F_{1,2(n-1),1-\alpha} \,\Big|\, \lambda \,\Big]. \tag{12}$$

This is computable with `stats::pf` and its `ncp` argument. For sample-size calculation, $n = 4\sigma_e^2\lambda/(T\Delta_0^2)$.

### 3.3.2 Assumptions of the closed form and which the study exercises

The closed form assumes (i) a categorical biomarker at the dichotomized cut-point with equal strata, (ii) sphericity of any within-participant factor with more than two levels, (iii) no carryover, (iv) a balanced design with no missing data, and (v) equal time spacing if time is kept as a factor. We state which of these the present study exercises.

Only assumption (i) is exercised. The executed RM-ANOVA arm pre-averages each phase to one value, so (ii) holds by construction. The DGP has no carryover, so (iii) holds. There is no dropout, so (iv) holds. Time is not a factor in the pre-averaged arm, so (v) does not arise. The comparison between strict RM-ANOVA and the continuous-biomarker contrast therefore isolates the dichotomization cost. The MMRM literature's handling of missing data[18] and the carryover specifications of the companion paper[36] address assumptions (iii) and (iv) in other studies. They are not tested here.

# 4 Simulation design

The study follows the ADEMP structure of[1]. It is exploratory. Its findings should be confirmed in an independent study.

## 4.1 Aims

The aims are (a) to separate the power cost of dichotomizing the biomarker from the power cost of the test procedure, and (b) to tabulate the size, power, coverage, and calibration ratio of the continuous-biomarker procedures at $N \in \{25, 35, 70, 100\}$.

## 4.2 Trial design

The design is a two-path open-label plus blinded discontinuation (OL+BDC) schedule. Participants take the drug open-label for 16 weeks with visits at weeks 4, 8, 12, and 16. A blinded phase follows with weekly visits at weeks 17 to 20. Each participant is allocated to one of two paths that differ in the week of blinded discontinuation. Table 1 gives the schedule. The baseline visit at week 0 is off drug with expectancy 0. Expectancy is 1 at the open-label visits and 0.5 at the blinded visits.

Table 1: The executed two-path OL+BDC schedule. "On" and "off" give the drug state at each visit. $e$ is the design expectancy, the participant's probability of being on drug by design.

| Visit | Week | Phase | $e$ | Path A | Path B |
|---|---|---|---|---|---|
| BL | 0 | baseline | 0 | off | off |
| 1 | 4 | open-label | 1 | on | on |
| 2 | 8 | open-label | 1 | on | on |
| 3 | 12 | open-label | 1 | on | on |

| Visit | Week | Phase | $e$ | Path A | Path B |
|---|---|---|---|---|---|
| 4 | 16 | open-label | 1 | on | on |
| 5 | 17 | blinded | 0.5 | on | on |
| 6 | 18 | blinded | 0.5 | on | off |
| 7 | 19 | blinded | 0.5 | off | off |
| 8 | 20 | blinded | 0.5 | off | off |

The total $N$ is split across the two paths as $\lceil N/2 \rceil$ and $\lfloor N/2 \rfloor$. Path A contributes 6 on-drug and 3 off-drug visits per participant (counting baseline), path B 5 and 4. This design is the two-path OL+BDC of the series' notation, not the four-path Hybrid design of[31], which adds a crossover block. The working protocol specified the Hybrid design; Table 2 records the change.

## 4.3 Data-generating mechanism

Data are generated by the pmsimstats package function `generateData()` with `dgp_architecture = 'mean_moderation'`. The outcome at each visit is a baseline level plus three components: a natural-history component, a placebo-response component, and a drug-response component. Each component follows a modified Gompertz curve in time on drug with the package's prazosin reference parameters. Within each component the visit-to-visit correlation is AR(1) with $\rho = 0.7$. The cross-component correlation is 0.1 at the same visit and 0.05 across visits (the package parameters `c.cf1t` and `c.cfct`). The biomarker is standard normal. Under the mean-shift encoding, the drug-response mean at on-drug visits is shifted by $\beta_{bm}\, b_i\, \sigma_{BR}$, where $b_i$ is the standardized biomarker and $\sigma_{BR} = 5$. The interaction on the outcome scale is therefore $\beta_{bm}\sigma_{BR}$, with an analytic value of 2.25 at $\beta_{bm} = 0.45$; a Monte Carlo reference fit at $N = 2000$ gave $-2.21$ in the sign convention of the fitted coefficient.

This configuration is the package default under the mean-shift encoding. It is neither the original DGP of[31] (compound symmetry within components, $\rho_{cs} = 0.8$) nor the extended DGP of the series' notation, which sets the cross-component correlation $c_1 = 0.2$. The carryover half-life is zero: the drug-response mean drops to zero at discontinuation. There is no dropout. Both paths' covariance matrices are positive definite at these parameters (minimum eigenvalue 0.168), and $\beta_{bm}$ does not enter the covariance under the mean-shift encoding. The stored run therefore did not rely on the package's former repair of non-positive-definite matrices, which is now disabled by default.

## 4.4 Estimands

The primary estimand is the interaction coefficient $\beta_{bm:D}$ on the outcome scale for the four continuous-biomarker arms. The strict RM-ANOVA arm targets a dichotomized-stratum contrast on a different scale and is excluded from the coefficient-based measures.

## 4.5 Methods

Six procedures were run. Each procedure is applied to the same replicate data set.

1. **Strict RM-ANOVA (dichotomized).** The biomarker is split at the sample median. Each participant's outcome is averaged within the on-drug and off-drug phases, baseline counted as off drug. A split-plot ANOVA (`aov` with `Error(ptID/Db)`) tests the stratum-by-phase interaction. This is the $2 \times 2$ form of equation (7).
2. **Continuous-biomarker contrast.** Each participant's on-drug mean minus off-drug mean, $\Delta_l$, is regressed on the continuous biomarker by ordinary least squares. The slope is the interaction estimate and its $t$-test on $N-2$ degrees of freedom is the test. This is the paired-difference statistic of the companion papers (their E9b, baseline counted as off drug).
3. **LME, model-based inference.** The package function `lme_analysis()` fits `Sx ~ bm + t + Dbc + bm:Dbc + De` by `nlme::lme` with a participant random intercept and `corCAR1(form = ~t | ptID)` residual correlation on calendar weeks. `t` is calendar week. `De` is the design expectancy. `Dbc` is the exposure predictor at an analysis half-life of zero, which is the binary on-drug indicator (0 at baseline). The test is the `nlme` Wald $t$ on `bm:Dbc` with the model-based standard error and `nlme`'s within-participant degrees of freedom (556 at $N = 70$). Apart from the `De` term, this is specification G1 of the companion paper[36], its Table of analysis codes, fitted at the true half-life.
4. **LME plus CR2.** The same fit as procedure 3. The standard error of `bm:Dbc` is replaced by the CR2 bias-reduced cluster-robust sandwich with Satterthwaite degrees of freedom[43,44], computed by `clubSandwich::coef_test` with participant as the cluster. This is the inference of G6 in the companion paper and of the calibration manuscript[37]. The point estimate is unchanged.
5. **GEE, Liang-Zeger sandwich.** `geepack::geeglm` fits `Sx ~ bm + t + Dbc + bm:Dbc` with a Gaussian family and `corstr = 'ar1'`. The working correlation is a discrete-lag AR(1) on observation order within participant. The visits are at weeks 0, 4, 8, 12, 16, 17, 18, 19, and 20, so the working correlation treats unequally spaced occasions as equally spaced. The LME arms use the continuous-time form instead. The test is the Wald $z$ with the robust sandwich of[19].
6. **GEE, Mancl-DeRouen sandwich.** The same fit as procedure 5. The sandwich meat is bias-corrected by inflating each cluster's residual as $(I - H_{ii})^{-1} r_i$[20], computed inline. The Wald statistic is referred to a $t$ distribution on $K - p$ degrees of freedom, with $K = N$ clusters and $p = 5$ fixed effects, the pairing[25] recommend. The stored 2026-10-08 run used a normal reference; the $t$-reference results below were recomputed from the stored per-replicate estimates and standard errors without refitting (`02-mancl-t-reference.R`). The inline sandwich was benchmarked against `geesmv::GEE.var.md` on 25 probe replicates at $N = 35$, $\beta_{bm} = 0.45$: the inline-to-`geesmv` ratio of the interaction standard error had mean 1.024, SD 0.021, and range 0.999 to 1.092.

The arms differ in their mean model as well as in their covariance treatment. The contrast

arm uses no time trend and no expectancy term. The GEE arms include `t`. The LME arms include `t` and `De`. The step from the contrast to the LME therefore changes the mean model as well as the covariance treatment, and we do not attribute it to the covariance treatment alone.

### 4.6 Performance measures

Power is $\Pr(p < 0.05)$ at $\beta_{bm} = 0.45$. Type I error is the same rate at $\beta_{bm} = 0$. Both carry the binomial MCSE $\sqrt{\hat{\pi}(1-\hat{\pi})/n_{\text{reps}}}$. For the coefficient-bearing arms we report bias relative to the Monte Carlo reference value, the empirical SE (the SD of the estimates), the mean model-based SE, the calibration ratio $\kappa$ (mean model SE over empirical SE; $\kappa > 1$ indicates a conservative test), and the coverage of the nominal 95 percent Wald interval. Every null cell is tested against 0.05 by an exact binomial test.

### 4.7 Deviations from the working protocol

The working protocol (`analysis/scripts/test-procedure-design-sensitivity/00-ademp-protocol.md`) describes three studies. The present paper executes a revised Study 1. Table 2 records every change with its rationale.

Table 2: Working protocol versus executed Study 1.

| Factor | Working protocol | Executed | Rationale |
|---|---|---|---|
| Design | Four-path Hybrid (8-week OL, 4-week BDC, two 4-week crossover blocks) | Two-path OL+BDC, visits at weeks 4, 8, 12, 16, 17, 18, 19, 20 (Table 1) | The package's `buildtrialdesign()` two-path schedule; the crossover block is deferred with the design grid |
| Substrate | `implementations/simple` | pmsimstats package (`generateData()`, `lme_analysis()`) | The package DGP is the one the companion papers use |
| RM-ANOVA form | Greenhouse-Geisser-corrected $F$ on the multi-timepoint series | Pre-averaged $2 \times 2$ split-plot $F$ (exact) | Isolates dichotomization; the GG-corrected multi-level form is a different procedure |

| Factor | Working protocol | Executed | Rationale |
|---|---|---|---|
| Test procedures | 3 (RM-ANOVA, LME, GEE-MD) | 6 (adds the contrast, Liang-Zeger GEE, LME plus CR2) | Contrast isolates dichotomization; Liang-Zeger documents the uncorrected GEE; CR2 is the series' inference |
| GEE reference | Not stated | Mancl-DeRouen with $t_{K-5}$; the normal reference is reported for comparison | [25] |
| Biomarker effect | $\{0, 0.3, 0.6\}$ | $\{0, 0.45\}$ | The series' reference effect size |
| Sample size $N$ | $\{35, 70, 100\}$ | $\{25, 35, 70, 100\}$ | Adds the smallest trial-relevant $N$ |
| Replicates | 1000 power, 5000 null | 1000 power, 2000 null | Run time; MCSE on a 5 percent rate is 0.5 points |
| Carryover half-life | 3 days | 0 (none) | Removes carryover from the dichotomization comparison; a carryover cell is future work |
| Seed mechanism | Per-replicate seed from a cell-descriptor hash | L'Ecuyer-CMRG streams, one per `mclapply` worker | `parallel` default; see the Reproducibility section |

**Units.** The protocol quotes half-lives in days and the interaction manuscripts of the series in weeks. One week is 7 days.

## 4.8 Execution

The 40 stored cells (5 procedures, 2 effect sizes, 4 sample sizes) ran in 587 s on 7 cores with `parallel::mclapply`. Convergence was 100 percent across the 60,000 stored fits. Per-replicate output is at `analysis/data/quick-sim/08-test-replicates.rds` and the cell-level summary at `08-test-summary.txt`. With 1000 replicates the MCSE on a power near 0.5 is 0.016. With 2000 replicates the MCSE on a rate near 0.05 is 0.005.

# 5 Results

## 5.1 Type I error and power

We applied the five stored procedures and the recomputed Mancl-DeRouen $t$ reference to every replicate. Table 3 gives the null rejection rates and Table 4 the power.

Table 3: Type I error rate (percent ± MCSE) under $H_0 : \beta_{bm:D} = 0$, 2000 replicates per cell. Nominal level 5 percent. An asterisk marks a cell whose exact binomial test against 0.05 has $p < 0.05$.

| Procedure | $N = 25$ | $N = 35$ | $N = 70$ | $N = 100$ |
|---|---|---|---|---|
| Strict RM-ANOVA (dichotomized) | 4.55 ± 0.47 | 4.15 ± 0.45 | 5.95 ± 0.53 | 5.20 ± 0.50 |
| Contrast (continuous biomarker) | 5.25 ± 0.50 | 4.50 ± 0.46 | 6.45 ± 0.55* | 4.60 ± 0.47 |
| LME, model-based (`corCAR1`) | 2.75 ± 0.37* | 2.05 ± 0.32* | 3.35 ± 0.40* | 2.90 ± 0.38* |
| GEE, Liang-Zeger sandwich | 9.70 ± 0.66* | 8.55 ± 0.63* | 7.90 ± 0.60* | 7.40 ± 0.59* |
| GEE, Mancl-DeRouen, $t_{K-5}$ | 4.95 ± 0.49 | 4.70 ± 0.47 | 5.85 ± 0.52 | 5.55 ± 0.51 |
| GEE, Mancl-DeRouen, normal | 6.35 ± 0.55* | 5.30 ± 0.50 | 6.25 ± 0.54* | 5.90 ± 0.53 |

Table 4: Power (percent ± MCSE) at $\beta_{bm} = 0.45$, 1000 replicates per cell.

| Procedure | $N = 25$ | $N = 35$ | $N = 70$ | $N = 100$ |
|---|---|---|---|---|
| Strict RM-ANOVA (dichotomized) | 35.8 ± 1.52 | 46.8 ± 1.58 | 75.3 ± 1.36 | 91.3 ± 0.89 |
| Contrast (continuous biomarker) | 53.8 ± 1.58 | 67.7 ± 1.48 | 91.5 ± 0.88 | 98.8 ± 0.34 |
| LME, model-based (`corCAR1`) | 56.5 ± 1.57 | 71.2 ± 1.43 | 94.5 ± 0.72 | 99.2 ± 0.28 |

| Procedure | $N = 25$ | $N = 35$ | $N = 70$ | $N = 100$ |
|---|---|---|---|---|
| GEE, Liang-Zeger sandwich | 67.2 ± 1.48 | 75.9 ± 1.35 | 94.5 ± 0.72 | 99.2 ± 0.28 |
| GEE, Mancl-DeRouen, $t_{K-5}$ | 52.0 ± 1.58 | 65.9 ± 1.50 | 92.2 ± 0.85 | 99.1 ± 0.30 |
| GEE, Mancl-DeRouen, normal | 57.3 ± 1.56 | 69.1 ± 1.46 | 92.7 ± 0.82 | 99.1 ± 0.30 |

## 5.2 The cost of dichotomization

The power gap between strict RM-ANOVA and the LME is mostly the cost of dichotomizing the biomarker. At $N = 25$ strict RM-ANOVA had power 0.358. The contrast, which differs only in keeping the biomarker continuous, had power 0.538. Dichotomization therefore cost 0.180. The LME had power 0.565, so the test procedure added 0.027. The decomposition is stable across $N$. The dichotomization cost was 0.209 at $N = 35$ (0.468 to 0.677), 0.162 at $N = 70$ (0.753 to 0.915), and 0.075 at $N = 100$ (0.913 to 0.988). The contrast-to-LME increment never exceeded 0.04 with model-based inference. With CR2 inference it was 0.035 to 0.065, the largest at $N = 35$ (0.677 to 0.742).

The observed cost agrees with the classical prediction. A median split of a normal predictor retains a fraction $2/\pi \approx 0.64$ of the squared correlation[13], so the non-centrality of the contrast test is multiplied by $\sqrt{2/\pi} \approx 0.80$. Solving the observed contrast power for its non-centrality at $N - 2$ degrees of freedom and scaling by 0.80 predicts dichotomized power of 0.375, 0.488, 0.758, and 0.920 at $N = 25, 35, 70, 100$. The observed values were 0.358, 0.468, 0.753, and 0.913, each within 1.3 MCSE of the prediction. The dichotomization result is therefore Cohen's loss observed in the N-of-1 interaction setting, not a new finding.

## 5.3 GEE calibration

The Liang-Zeger GEE over-rejected at every $N$. Its null rate was 0.097 at $N = 25$ and fell only to 0.086, 0.079, and 0.074 at $N = 35, 70, 100$ (exact binomial $p < 10^{-5}$ in every cell). The inflation persists at trial-relevant sizes.

The Mancl-DeRouen correction with the normal reference did not restore nominal size. Its null rates were 0.064, 0.053, 0.063, and 0.059. The $N = 25$ cell is 2.8 MCSE above nominal (binomial $p = 0.008$) and the $N = 70$ cell 2.6 MCSE above (binomial $p = 0.014$). This is the residual over-rejection that[25] report for a bias-corrected sandwich with a Wald $z$ test at small cluster counts.

The same correction with a $t$ reference on $K - 5$ degrees of freedom (20, 30, 65, and 95) gave null rates of 0.050, 0.047, 0.059, and 0.056. No cell differs from 0.05 by the binomial test (smallest $p = 0.09$, at $N = 70$). Its coverage of the true interaction at $\beta_{bm} = 0.45$ was 95.6,

95.0, 95.4, and 94.4 percent. The power cost relative to the Liang-Zeger GEE was 15 points at $N = 25$ (0.672 to 0.520), 10 points at $N = 35$ (0.759 to 0.659), 2 points at $N = 70$, and none at $N = 100$. The Liang-Zeger GEE's apparent power advantage over the LME at small $N$ is the residue of its over-rejection, not an efficiency gain. With the $t$ reference the corrected GEE sits 0.00 to 0.05 below the model-based LME and within 0.02 of the contrast.

### 5.4 Standard-error calibration

Table 5 gives the calibration ratio $\kappa$ for every coefficient-bearing arm and cell. Table 6 gives coverage and bias.

Table 5: Calibration ratio $\kappa$ (mean model-based SE divided by the empirical SD of the estimates) by arm, effect size and $N$. Values above 1 indicate a conservative test; values below 1 an anti-conservative one.

| Arm | $\beta_{bm}$ | $N = 25$ | $N = 35$ | $N = 70$ | $N = 100$ |
|---|---|---|---|---|---|
| Contrast | 0 | 0.98 | 0.98 | 0.96 | 1.02 |
| Contrast | 0.45 | 0.99 | 1.00 | 0.94 | 0.98 |
| LME, model-based | 0 | 1.13 | 1.13 | 1.10 | 1.15 |
| LME, model-based | 0.45 | 1.16 | 1.15 | 1.11 | 1.11 |
| GEE, Liang-Zeger | 0 | 0.88 | 0.91 | 0.94 | 0.97 |
| GEE, Liang-Zeger | 0.45 | 0.91 | 0.93 | 0.96 | 0.98 |
| GEE, Mancl-DeRouen | 0 | 1.03 | 1.02 | 1.00 | 1.01 |
| GEE, Mancl-DeRouen | 0.45 | 1.07 | 1.05 | 1.01 | 1.02 |

Table 6: Coverage (percent) of the nominal 95 percent interval at $\beta_{bm} = 0.45$, and the largest relative bias of $\widehat{\beta}_{bm:D}$ across all cells. The Monte Carlo reference value is $-2.21$. The Mancl-DeRouen row uses the $t_{K-5}$ interval; the normal-reference interval covered 94.3, 94.2, 95.0 and 94.0 percent.

| Arm | $N = 25$ | $N = 35$ | $N = 70$ | $N = 100$ | Max relative bias |
|---|---|---|---|---|---|
| Contrast | 94.9 | 95.0 | 93.5 | 94.8 | 3.0% |
| LME, model-based | 97.7 | 97.3 | 97.3 | 97.0 | 2.6% |
| GEE, Liang-Zeger | 90.0 | 91.5 | 94.1 | 93.3 | 3.8% |
| GEE, Mancl-DeRouen, $t_{K-5}$ | 95.6 | 95.0 | 95.4 | 94.4 | 3.8% |

The relative bias of $\widehat{\beta}_{bm:D}$ was below 4 percent of the reference value in every cell for every coefficient-bearing arm. The differences in size, power, and coverage are therefore variance-estimation effects.

The contrast is calibrated: $\kappa$ is 0.94 to 1.02 and coverage 93.5 to 95.0 percent. Its null rate at $N = 70$ (0.0645, binomial $p = 0.004$) is the one excursion among its four null cells. With 20 null cells in the study one such excursion is expected, and the arm's other three null cells are at nominal.

The Liang-Zeger GEE under-estimates its standard error: $\kappa$ is 0.88 to 0.97 under the null and 0.91 to 0.98 at $\beta_{bm} = 0.45$, with the largest deficits (9 and 7 percent at the power cells for $N = 25$ and 35) at the smallest $N$. Its coverage is 90.0 percent at $N = 25$. The Mancl-DeRouen correction brings $\kappa$ to 1.00 to 1.07.

## 5.5 The conservatism of the model-based LME

The LME's null rejection rate was 0.0275, 0.0205, 0.0335, and 0.0290. Every cell is below 0.05 by 3.4 to 6.1 MCSE (binomial $p < 0.001$ in each). Its coverage is 97.0 to 97.7 percent and its $\kappa$ is 1.10 to 1.16. The model-based standard error exceeds the empirical standard error by 10 to 16 percent in every cell.

This is a working-covariance calibration defect, not a degrees-of-freedom effect. The fitted covariance (random intercept plus continuous-time AR(1)) is applied to data generated from three AR(1) components with distinct curves, a separate baseline level, and cross-component correlations. The `nlme` Wald test assigns the within-participant term `bm:Dbc` the inner-level degrees of freedom, 556 at $N = 70$, so no choice of reference distribution can produce a 2 percent size. A Kenward-Roger[45] or Satterthwaite adjustment of the model-based test can only reduce the rejection rate further. Both inflate the estimated variance or reduce the denominator degrees of freedom. They are remedies for anti-conservative tests, not conservative ones.

The companion papers diagnose and remedy this defect. The calibration manuscript[37] traces the over-estimated standard error to the working covariance. The carryover manuscript[36] replaces the model-based standard error by the CR2 cluster-robust sandwich with Satterthwaite degrees of freedom (its rung G6), which moves the null rejection rate from the conservative G1 value to near nominal. Procedure 4 (Section 4.5) applies that inference to the fit evaluated here.

## 5.6 LME plus CR2 arm

Table 7: LME plus CR2 (Satterthwaite) arm: Type I error at $\beta_{bm} = 0$ (2000 replicates), and power, coverage and $\kappa$ at $\beta_{bm} = 0.45$ (1000 replicates). Values are percent $\pm$ MCSE where applicable.

| Quantity | $N = 25$ | $N = 35$ | $N = 70$ | $N = 100$ |
|---|---|---|---|---|
| Type I error (percent) | 5.05 $\pm$ 0.49 | 4.30 $\pm$ 0.45 | 5.90 $\pm$ 0.53 | 5.15 $\pm$ 0.49 |
| Power (percent) | 57.3 $\pm$ 1.56 | 74.2 $\pm$ 1.38 | 96.3 $\pm$ 0.60 | 99.7 $\pm$ 0.17 |
| Coverage (percent) | 95.5 | 94.2 | 93.9 | 94.6 |
| $\kappa$ | 0.98 | 0.98 | 0.96 | 0.96 |

Table 7 reports the arm. Its $\kappa$ and null rejection rate should be read against the model-based LME row of Tables 3 and 5. The point estimates are identical, so the two arms differ only in the standard error. CR2 brings $\kappa$ from 1.10 to 1.16 down to 0.96 to 0.98. Its null rejection rate

is 4.3 to 5.9 percent, each within 2 MCSE of 5 percent. Its power exceeds the model-based LME by 0.5 to 3.0 points at every $N$.

# 6 Recommendation

The recommendation is scoped to what was tested: a two-path OL+BDC design, no carry-over, no dropout, the mean-shift encoding of the interaction, and the procedures of Section 4.5. The overriding recommendation is design-agnostic. Keep the biomarker continuous. Dichotomizing it forfeits 0.08 to 0.21 of absolute power here, which is the classical $2/\pi$ loss.

| Scenario | Preferred procedure | Rationale |
|---|---|---|
| Any $N$, continuous biomarker, inference to match the series | LME with CR2 Satterthwaite inference (procedure 4) | The inference of the companion papers' G6 to G9; corrects the model-based LME's conservatism; size 0.043 to 0.059, $\kappa$ 0.96 to 0.98, the highest power of the continuous-biomarker arms with nominal size |
| Any $N$, model-based LME only | LME, model-based (`corCAR1`) | Power 0.5 to 3.0 points below the CR2 arm and conservative (size 0.02 to 0.03, $\kappa$ 1.10 to 1.16); do not apply Kenward-Roger or Satterthwaite to the model-based test, which would lower the size further |
| Transparent classical analysis | Continuous-biomarker contrast | Within 0.04 of the model-based LME and 0.07 of the CR2 arm on power; size at nominal in 3 of 4 cells; $\kappa$ 0.94 to 1.02; no covariance modeling |
| Marginal (GEE) inference required | GEE, Mancl-DeRouen sandwich with $t_{K-p}$ | Size 0.047 to 0.059, none distinguishable from 0.05; coverage 94 to 96 percent; power 0.00 to 0.05 below the model-based LME |
| Reviewer-mandated classical analysis | Strict RM-ANOVA | Report alongside a continuous-biomarker analysis with the decomposition |

| Scenario | Preferred procedure | Rationale |
|---|---|---|
| GEE with the Liang-Zeger sandwich, or Mancl-DeRouen with a normal reference | Not recommended | Size 0.074 to 0.097 (Liang-Zeger); 0.053 to 0.064 (Mancl-DeRouen, normal) |

The recommendation for a trial with carryover is not made here. The companion paper[36] recommends G9, the exposure-decay predictor with the half-life chosen by AIC and CR2 inference, for the Hybrid design under the correlation encoding, with F1 (expectancy terms, fixed half-life, CR2) as the fallback. The present study has no carryover and does not exercise the half-life question. Its LME arm shares G1's formula, not G9's. The "preferred default" of an earlier draft of this paper, the model-based LME, is the analysis the companion papers replace. We retain it in the table as the stored comparator and label it as such.

# 7 Discussion

## 7.1 What the study settles

The study settles four points at its resolution. First, the power gap between strict RM-ANOVA and the LME is the cost of dichotomization. The continuous-biomarker contrast recovers 0.180 of the 0.207 gap at $N = 25$ and the pattern holds across the grid. The loss matches Cohen's $2/\pi$ factor within MCSE. Second, the Liang-Zeger GEE over-rejects at every $N$ examined, and the Mancl-DeRouen correction restores size only when paired with a $t$ reference on $K - p$ degrees of freedom. Third, the four continuous-biomarker arms are unbiased, so their differences are variance-estimation effects that the $\kappa$ table explains. Fourth, the model-based LME is conservative because its working covariance over-estimates the standard error by 10 to 16 percent. A degrees-of-freedom correction would move it further from nominal. The CR2 cluster-robust standard error is the remedy. Here it brings the null rejection rate to 4.3 to 5.9 percent and raises power at every $N$.

## 7.2 Limitations

The study has one design, one effect size, no carryover, and no dropout. The GEE working correlation is a discrete-lag AR(1) on unequally spaced visits, which mismatches the DGP and the LME's continuous-time form; a closer working correlation might change the GEE power by a few points. The LME and GEE arms include a time trend and the LME an expectancy term, while the contrast includes neither, so the contrast-to-LME step is not a pure covariance comparison. The Mancl-DeRouen $t$-reference numbers were recomputed from stored estimates and then reproduced exactly by a full rerun that added the CR2 arm. The rerun reproduced every stored arm to machine precision. The study is exploratory.

### 7.3 Future work

Two extensions are recorded. A carryover cell at $t_{1/2} = 0.5$ week, the series' standard grid value, would exercise the no-carryover assumption rather than assert it. A cycle-by-period design grid (cycle count, on- and off-drug period lengths, on/off symmetry, run-in) was planned as Studies 2 and 3 of the working protocol and is not delivered here. The recommendation in Section 6 holds at the fixed design only.

## 8 Conclusions

Keep the biomarker continuous. Dichotomization cost 0.08 to 0.21 of absolute power across $N \in \{25, 35, 70, 100\}$, which is the classical $2/\pi$ loss, while the test procedure moved power by at most 0.07 among procedures with nominal or conservative size. The model-based LME was conservative, with size 0.02 to 0.03 and a model standard error 10 to 16 percent above the empirical one. That defect is in the working covariance, not in the degrees of freedom. The series' CR2 cluster-robust inference repairs it: the same fit rejected at 0.043 to 0.059 under the null and had the highest power of the continuous-biomarker arms with nominal size. The Liang-Zeger GEE over-rejected at every $N$. The Mancl-DeRouen sandwich with a $t_{K-p}$ reference held size at 0.047 to 0.059 with 94 to 96 percent coverage, at a power cost of 10 to 15 points relative to the uncorrected GEE at $N \leq 35$. The strict RM-ANOVA $F$-test is a two-sample $t$-test on within-participant phase differences, exact for any within-participant covariance once each phase is averaged, and its only cost here is the dichotomization.

## 9 Glossary

**Aggregated N-of-1 trial.** Many single-participant on/off series pooled into one model.

**OL+BDC.** Open-label titration followed by blinded discontinuation. The two-path version used here is defined in Table 1.

**Hybrid design.** The[31] design: open-label titration, blinded discontinuation, then a brief crossover. Not used here.

**Mean-shift encoding.** The interaction is a shift of the drug-response mean by the centered biomarker. The data value is `mean_moderation`.

**Correlation encoding.** The interaction is a correlation between the biomarker and the drug response that is present on drug and absent off drug. The companion papers' $c_{bm}$. Not used here.

**$\beta_{bm}$.** The interaction multiplier on the standardized biomarker under the mean-shift encoding. The outcome-scale interaction is $\beta_{bm}\sigma_{BR}$.

**$\beta_{bm:D}$.** The interaction coefficient on the outcome scale, the coefficient of `bm:Dbc` in the LME and GEE fits.

`Dbc`. The exposure predictor. At an analysis half-life of zero it is the binary on-drug indicator, 0 at baseline.

`De`. The design expectancy at each visit (Table 1).

`corCAR1`. The continuous-time AR(1) residual correlation of `nlme`, $\varphi^{|w_t - w_s|}$ on calendar weeks.

**G1, G6, G9, F1.** Analysis codes of the companion paper[36]. G1 is the binary-indicator LME with model-based inference. G6 is G1 with CR2 inference. G9 is the exposure-decay predictor with the half-life chosen by AIC and CR2 inference. F1 adds expectancy terms at a fixed half-life with CR2.

**E9b.** The companion papers' label for the paired-difference statistic with baseline counted as an off-drug visit; the contrast arm here.

**CR2.** The bias-reduced cluster-robust variance estimator of[43] with Satterthwaite degrees of freedom.

**Liang-Zeger sandwich.** The robust variance estimator of[19]. Some GEE literature calls the model-based variance "naive"; this paper does not use that word.

**Mancl-DeRouen sandwich.** The bias-corrected sandwich of[20], with residuals inflated by $(I - H_{ii})^{-1}$.

$\kappa$**.** The calibration ratio: mean model-based standard error divided by the empirical standard deviation of the estimates. $\kappa > 1$ indicates a conservative test.

**Extended DGP, original DGP.** The series' names for the recommended data-generating process ($c_1 = 0.2$, decaying correlation, carryover from the last on-drug mean) and for the[31] process. The configuration used here is neither.

**MCSE.** Monte Carlo standard error.

# 10 Conflict of interest

The authors declare no conflicts of interest.

# 11 Funding

This work received no specific funding.

# 12 Data availability statement

The data and code that support the findings of this study are available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and cell-level summaries are under `analysis/data/`, and the drivers under `analysis/scripts/`. The exact paths are in the Reproducibility section.

# 13 Reproducibility

The stored simulation ran with R 4.6.x. The project ships a zzcollab Docker container (`Dockerfile`, `renv.lock`) for container-based reproduction. Packages used are `nlme`, `geepack`, `geesmv` (benchmark only), `clubSandwich` (the CR2 arm), `data.table`, `parallel`, `rmarkdown` and `bookdown`.

**Random-number generator.** The driver sets `RNGkind("L'Ecuyer-CMRG")` and `set.seed(20260507)`, then runs the cell-by-replicate job list through `parallel::mclapply` with `mc.set.seed = TRUE` and `mc.preschedule = TRUE`. Each forked worker receives one L'Ecuyer stream and processes a prescheduled block of jobs from it sequentially. Reproduction therefore requires the same worker count (7 for the stored run), the same job list, and the same platform. The reference run that pins the estimand (6 draws at $N = 2000$) precedes the job list and advances the generator, so it must also be run unchanged. Results from the Docker container and from a local session are not draw-for-draw identical (different BLAS).

**Data and code.** Per-replicate outputs are at `analysis/data/quick-sim/08-test-replicates.rds`; the cell-level summary at `08-test-summary.txt`; the Monte Carlo reference estimand at `08-test-betatrue.txt`; the `geesmv` benchmark at `08-test-mancl-benchmark.txt`; the recomputed Mancl-DeRouen $t$-reference summary, the binomial tests of the null cells, and the $\kappa$ table at `08-test-mancl-tref.txt`, `08-test-null-binomial.txt` and `08-test-kappa.txt`. The driver is `analysis/scripts/test-procedure-design-sensitivity/01-study1-test-procedure.R` and the recomputation script `02-mancl-t-reference.R` in the same directory. The working protocol is `00-ademp-protocol.md` there. The manuscript source, bibliography (`references.bib`) and CSL file are at `analysis/report/08-test-procedure-design-sensitivity/`.

# 14 References


1. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*. 2019;38(11):2074-2102. doi:10.1002/sim.8086

2. Kent DM, Paulus JK, Klaveren D van, et al. The Predictive Approaches to Treatment effect Heterogeneity (PATH) Statement. *Annals of Internal Medicine*. 2020;172(1):35-45. doi:10.7326/M18-3667

3. Kent DM, Klaveren D van, Paulus JK, et al. The PATH statement: Explanation and elaboration. *Annals of Internal Medicine*. 2020;172(1):W1-W25. doi:10.7326/M18-3668

4. Haverkamp N, Beauducel A. Violation of the sphericity assumption and its effect on type-I error rates in repeated measures ANOVA and multi-level linear models. *Frontiers in Psychology*. 2017;8:1841. doi:10.3389/fpsyg.2017.01841

611 5. Blanca MJ, Arnau J, García-Castro FJ, Alarcón R, Bono R. Repeated measures ANOVA and adjusted tests when sphericity is violated: Which procedure is best? *Frontiers in Psychology*. 2023;14:1192453. doi:10.3389/fpsyg.2023.1192453

612 6. Natesan Batley P, McClure EB, Brewer B, et al. Evidence and reporting standards in N-of-1 medical studies: A systematic review. *Translational Psychiatry*. 2023;13(1):263. doi:10.1038/s41398-023-02562-8

613 7. Stunnenberg BC, Berends J, Griggs RC, et al. N-of-1 trials in neurology: A systematic review. *Neurology*. 2022;98(2):e174-e185. doi:10.1212/WNL.0000000000012998

614 8. Shaffer JA, Kronish IM, Falzon L, Cheung YK, Davidson KW. N-of-1 randomized intervention trials in health psychology: A systematic review and methodology critique. *Annals of Behavioral Medicine*. 2018;52(9):731-742. doi:10.1093/abm/kax026

615 9. Defelippe VM, Thiel GJMW van, Otte WM, Schutgens REG, Stunnenberg B, et al. Toward responsible clinical n-of-1 strategies for rare diseases. *Drug Discovery Today*. 2023;28(10):103688. doi:10.1016/j.drudis.2023.103688

616 10. Vohra S, Shamseer L, Sampson M, et al. CONSORT extension for reporting N-of-1 trials (CENT) 2015 statement. *Journal of Clinical Epidemiology*. 2015;76:9-17. doi:10.1016/j.jclinepi.2015.05.004

617 11. Shamseer L, Sampson M, Bukutu C, Schmid CH, et al. CENT 2015: Explanation and elaboration. *Journal of Clinical Epidemiology*. 2015;76:18-46. doi:10.1016/j.jclinepi.2015.05.018

618 12. Kwasnicka D, Inauen J, Nieuwenboom W, et al. Challenges and solutions for N-of-1 design studies in health psychology. *Health Psychology Review*. 2019;13(2):163-178. doi:10.1080/17437199.2018.1564627

619 13. Cohen J. The cost of dichotomization. *Applied Psychological Measurement*. 1983;7(3):249-253. doi:10.1177/014662168300700301

620 14. MacCallum RC, Zhang S, Preacher KJ, Rucker DD. On the practice of dichotomization of quantitative variables. *Psychological Methods*. 2002;7(1):19-40. doi:10.1037/1082-989x.7.1.19

621 15. Royston P, Altman DG, Sauerbrei W. Dichotomizing continuous predictors in multiple regression: A bad idea. *Statistics in Medicine*. 2006;25(1):127-141. doi:10.1002/sim.2331

622 16. Ma Y, Mazumdar M, Memtsoudis SG. Beyond repeated-measures analysis of variance: Advanced statistical methods for the analysis of longitudinal data in anesthesia research. *Regional Anesthesia and Pain Medicine*. 2012;37(1):99-105. doi:10.1097/AAP.0b013e31823ebc74

623 17. Mallinckrodt CH, Clark SWS, Carroll RJ, Molenberghs G. Assessing response profiles from incomplete longitudinal clinical trial data under regulatory considerations. *Journal of Biopharmaceutical Statistics*. 2003;13(2):179-190. doi:10.1081/BIP-120019265

624 18. Mallinckrodt CH, Lane PW, Schnell D, Peng Y, Mancuso JP. Recommendations for the primary analysis of continuous endpoints in longitudinal clinical trials. *Drug Information Journal*. 2008;42(4):303-319. doi:10.1177/009286150804200402

625 19. Liang KY, Zeger SL. Longitudinal data analysis using generalized linear models. *Biometrika*. 1986;73(1):13-22. doi:10.1093/biomet/73.1.13

626 20. Mancl LA, DeRouen TA. A covariance estimator for GEE with improved small-sample properties. *Biometrics*. 2001;57(1):126-134. doi:10.1111/j.0006-341X.2001.00126.x

627 21. Kauermann G, Carroll RJ. A note on the efficiency of sandwich covariance matrix estimation. *Journal of the American Statistical Association*. 2001;96(456):1387-1396. doi:10.1198/016214501753382309

628 22. Fay MP, Graubard BI. Small-sample adjustments for Wald-type tests using sandwich estimators. *Biometrics*. 2001;57(4):1198-1206. doi:10.1111/j.0006-341X.2001.01198.x

629 23. Wang M, Kong L, Li Z, Zhang L. Covariance estimators for generalized estimating equations (GEE) in longitudinal analysis with small samples. *Statistics in Medicine*. 2016;35(10):1706-1721. doi:10.1002/sim.6817

630 24. Pan W, Wall MM. Small-sample adjustments in using the sandwich variance estimator in generalized estimating equations. *Statistics in Medicine*. 2002;21(10):1429-1441. doi:10.1002/sim.1142

631 25. Li P, Redden DT. Small sample performance of bias-corrected sandwich estimators for cluster-randomized trials with binary outcomes. *Statistics in Medicine*. 2015;34(2):281-296. doi:10.1002/sim.6344

632 26. Tong G, Nevins P, Ryan M, others, Taljaard M. A review of current practice in the design and analysis of extremely small stepped-wedge cluster randomized trials. *Clinical Trials*. 2025;22(1):45-56. doi:10.1177/17407745241276137

633 27. Hedges LV, Shadish WR, Natesan Batley P. Power analysis for single-case designs: Computations for $(AB)^k$ designs. *Behavior Research Methods*. 2023;55(7):3494-3503. doi:10.3758/s13428-022-01971-9

634 28. Bouwmeester S, Jongerling J. Power of a randomization test in a single case multiple baseline AB design. *PLoS ONE*. 2020;15(2):e0228355. doi:10.1371/journal.pone.0228355

635 29. Senn S. The AB/BA crossover: Past, present and future? *Statistical Methods in Medical Research*. 1994;3(4):303-324. doi:10.1177/096228029400300402

636 30. Senn S, Rolfe K, Julious SA. Investigating variability in patient response to treatment: A case study from a replicate cross-over study. *Statistical Methods in Medical Research*. 2011;20(6):657-666. doi:10.1177/0962280210379174

637 31. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: Statistical methods and theoretical findings. *Frontiers in Digital Health*. 2020;2:13. doi:10.3389/fdgth.2020.00013

638 32. Blackston JW, Chapple AG, McGree JM, McDonald S, Nikles J. Comparison of aggregated N-of-1 trials with parallel and crossover randomized controlled trials using simulation studies. *Healthcare*. 2019;7(4):137. doi:10.3390/healthcare7040137

639 33. Schork NJ, Beaulieu-Jones B, Liang WS, Smalley S, Goetz LH. Exploring human biology with N-of-1 clinical trials. *Cambridge Prisms: Precision Medicine*. 2023;1:e12. doi:10.1017/pcm.2022.15

640 34. Zucker DR, Ruthazer R, Schmid CH. Individual (N-of-1) trials can be combined to give population comparative treatment effect estimates: Methodologic considerations. *Journal of Clinical Epidemiology*. 2010;63(12):1312-1323. doi:10.1016/j.jclinepi.2010.04.020

641 35. Berger VW, Rezvani A, Makarewicz VA. Direct effect on validity of response run-in selection in clinical trials. *Controlled Clinical Trials*. 2003;24(2):156-166. doi:10.1016/s0197-2456(02)00316-1

642 36. pmsimstats team. Carryover representation and inference for biomarker-treatment interaction tests in aggregated N-of-1 trials under covariance moderation: A staged simulation study and a recommended analysis for the Hybrid design. Published online 2026.

643 37. pmsimstats team. Type I calibration of fixed-effect interaction tests in linear mixed models: A staged diagnosis of working-covariance misspecification and a cluster-robust remedy. Published online 2026.

644 38. Maxwell SE, Delaney HD, Kelley K. *Designing Experiments and Analyzing Data: A Model Comparison Perspective*. 3rd ed. Routledge; 2018.

645 39. Senn S. *Cross-over Trials in Clinical Research*. 2nd ed. Wiley; 2002.

646 40. Mauchly JW. Significance test for sphericity of a normal n-variate distribution. *Annals of Mathematical Statistics*. 1940;11(2):204-209. doi:10.1214/aoms/1177731915

647 41. Greenhouse SW, Geisser S. On methods in the analysis of profile data. *Psychometrika*. 1959;24(2):95-112. doi:10.1007/BF02289823

648 42. Huynh H, Feldt LS. Estimation of the Box correction for degrees of freedom from sample data in randomized block and split-plot designs. *Journal of Educational Statistics*. 1976;1(1):69-82. doi:10.3102/10769986001001069

649 43. Pustejovsky JE, Tipton E. Small-sample methods for cluster-robust variance estimation and hypothesis testing in fixed effects models. *Journal of Business and Economic Statistics*. 2018;36(4):672-683. doi:10.1080/07350015.2016.1247004

650 44. Satterthwaite FE. An approximate distribution of estimates of variance components. *Biometrics Bulletin*. 1946;2(6):110-114. doi:10.2307/3002019

651 45. Kenward MG, Roger JH. Small sample inference for fixed effects from restricted maximum likelihood. *Biometrics*. 1997;53(3):983-997. doi:10.2307/2533558